Boston Crime Analysis

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Introduction

Boston, MA is one of the United States' most iconic cities. Its rich history dates back to the time of the first European settlers, and it was the site of many meaningful events around the Revolutionary War, some of which have become the foundation of early American lore. Today it is home to several of the world's most prestigious universities, boasts a rabid sports culture, and is a burgeoning hub of the tech industry in America. It is the tenth largest metro area by population in the country, with over 4.6 million residents. With any city of size and industry, of course comes crime. We will explore the dynamics of crime in Boston and what population demographics and geographical factors may contribute to the city's crime. We are especially curious about the effects gentrification has on crime in Boston and the possibility of drawing conclusions regarding gentrification from observable shifts in crime statistics.

For quick context regarding crime in Boston - Boston's violent crime rate and property crime rate are 78% above and 12% below the national average respectively.[1] While Boston experiences higher than average crime rates in these two areas, Massachusetts as a whole is ranked as the tenth safest state in the US.[2] In addition, Boston crime falls in line with other cities of comparable size around the US.[3]

Statement of Goals

In this project, we try to investigate the crime situation in the Boston area: What is the general crime situation? Which crime types are the most common? Where did they occur? Are there patterns associated with crime ratios? Is there any trend in crime location/numbers over the years? We are also interested in learning about how the changes in crime locations associated with gentrification and average income in each district. Our intended audience of our analysis would be the Boston Police Department (BPD) or other government agencies or commercial security companies. By understanding the current situation and potential trend, they can more strategically deploy their staff and resources. Our analysis should also be helpful for anyone who

[1] http://www.city-data.com/crime/crime-Boston-Massachusetts.html
[2] https://www.usnews.com/news/best-states/slideshows/10-safest-states-in-america
[3] https://realestate.usnews.com/places/massachusetts/boston/crime

wants a deep-dived analysis of crime in Boston. Additionally the analysis can be extended to other areas in the future.

Data Cleaning

Our main dataset is sourced directly from the city of Boston's website.[4] The city provides a report for every incident to which the Boston Police Department responds, with each incident represented as a single row in the dataset. It covers every incident investigated within the city limits between June 15, 2015 and September 29, 2019, and includes information about the specific offense, with varying degrees of detail about the event, the location, and date and time. The dataset contains almost 437,000 rows, and is, not surprisingly, quite messy. Not every police call is the result of a crime, some calls produce multiple incident reports, and some other observations are simply entered incorrectly. Thus, cleaning our data was a lengthy process.

For example, on August 29, 2015, a shooting on Creston St. in the Roxbury neighborhood left one dead and two others injured.[5] This resulted in 20 individual incident reports. The reports included the expected incidents, homicide and aggravated assault, but also incidents for medical treatment administered, the ballistics investigation into the gun shots, and a missing person found, as apparently one of the victims had been previously reported missing. The latter three incidents are not actually crimes, so we aimed to wrangle our data in a way that would omit non-crime incidents. Further, each of the five incident reports was entered identically four times.
We removed duplicate rows, however this was a difficult decision, as we were not able to definitively determine how BPD reports multiple similar crimes. For example, it is unclear if a double murder would produce a single incident report for two murders, two incident reports with the same incident number, one for each murder in a single incident, or two incident reports with different incident numbers, one incident for each murder. With no detailed information about the reporting methods provided by the city, and based on what we observed in the data, we believe inconsistencies in reporting protocol exist. Accordingly, we eliminate all duplicate incidents, as there are assuredly more instances of double reporting than double murders.

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[4] https://data.boston.gov/dataset/crime-incident-reports-august-2015-to-date-source-new-system

[5] https://bpdnews.com/news/2015/8/30/update-victim-identified-in-death-investigation-in-the-area-of-29-creston-street-in-dorchester

The number of duplicate entries removed was around 750. While that may seem like an insignificant number in aggregate from a dataset of 430,000, it did impact certain subsets and groupings of the data. In the half of 2015 data reported, duplicates accounted for 13 of the 40 homicide reports, which would have increased the reported murder rate by in Boston by nearly 50%.

We did not initially identify duplicate rows as a symptom of dirty data. In fact, we began searching for duplicate incident numbers as a way of identifying major crimes. Our initial interpretation of the aforementioned shooting with 20 incident reports was of a bloody rampage with four people killed. A search for news articles about the assumed massacre yielded the truth.

In addition to dropping duplicates, we dropped approximately 2,250 rows that did not have district information. Next, we began determining what incidents were relevant to our scope of crime analysis.

Discerning between a crime and an incident became a major focus of our data cleaning, and we quickly realized many of the incident reports did not reflect a real crime. Examples of reports that do not criminal behavior include the offense code group of "Search Warrants", "Service", or "Towed." Further, incidents for "Auto Theft Recovery" and "Recovered Stolen Property," while relating to a crime, are not crimes, and would result in double reporting the initial auto theft or burglary. We identified 26 incident types from the offense code group column and five incident types from the offense description to classify as “non-crimes,” removing some 205,250 rows from the set.

The cleaning process was tedious and ongoing. Some problems, like dropping rows without district information, were obvious and easy to fix, but others, like identifying and dropping duplicates did not immediately present themselves, and we did not discover them until well into the exploratory process.

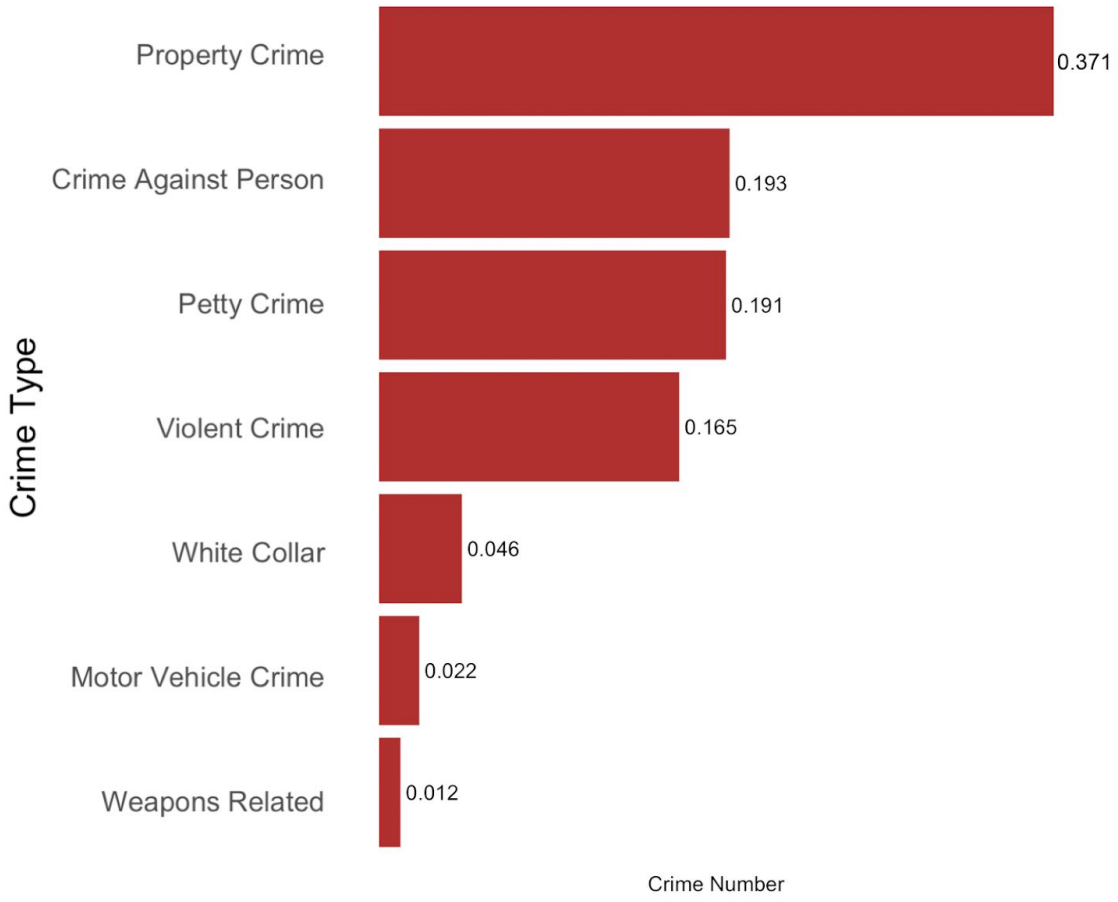
With the data cleaned, we still had over 60 individual incident types to sift through, and we turned our attention to grouping them into buckets.

The United States Department of Justice justice defines violent crime as instances of murder, non-negligible manslaughter, rape, sexual assault, aggravated assault, simple assault, and robbery. Boston Police report similar crimes, manslaughter, homicide, robbery, aggravated assault, and simple assault, which were straight forward for grouping. Strangely, there was not a single report that included an offense code or offense description referencing any sexual violence. Our best guess is that, for whatever reason, these crimes are being labeled as some other offense, most likely one of the assault categories, however it is possible they have been removed from the dataset entirely. As noted, each report comes with a latitude and longitude of the incident, which could create potential privacy issues for victims of sex crimes.

Property crime is another official crime group tracked by government agencies like the D.O.J. and the F.B.I. Per the F.B.I., property crime includes burglary, larceny-theft, motor vehicle theft, and arson. Boston Police breaks some of these categories down further, noting the kind of burglary and larceny or the setting of the crime.

For the remaining groupings we had to use our best judgement and determine similarities ourselves. Offense codes for explosives, firearm violations, bomb hoax were grouped into non-violent “weapons related” crimes. Operating under the influence and license plate related incidents were grouped into “motor vehicle crime.” Offenses like evading fare, disorderly conduct, and drug offenses fell into “petty crime.” Fraud, counterfeiting, and similar offenses were placed in the “white collar” group. A more sprawling category we named “crimes against a person” included non-violent offenses like criminal harassment, restraining order violations, and threats to do bodily harm.

In total we had seven crime groups that will be discussed in detail:



Our project used two datasets, Boston Crime Data and Boston Demographics Data. For Boston Crime Data, we first cleaned the column names, and then changed the district codes into district names for clarity. For the Boston Demographics Data, we want to know the population and per capita income in each district. We also joined the education data from this data set as well as the race by neighborhood data. One difficulty we encountered is that the district category here is different from the police districts in Boston Crime Data, so we found the Wikipedia of Boston Police Department and combined the 23 districts into 11 police districts.

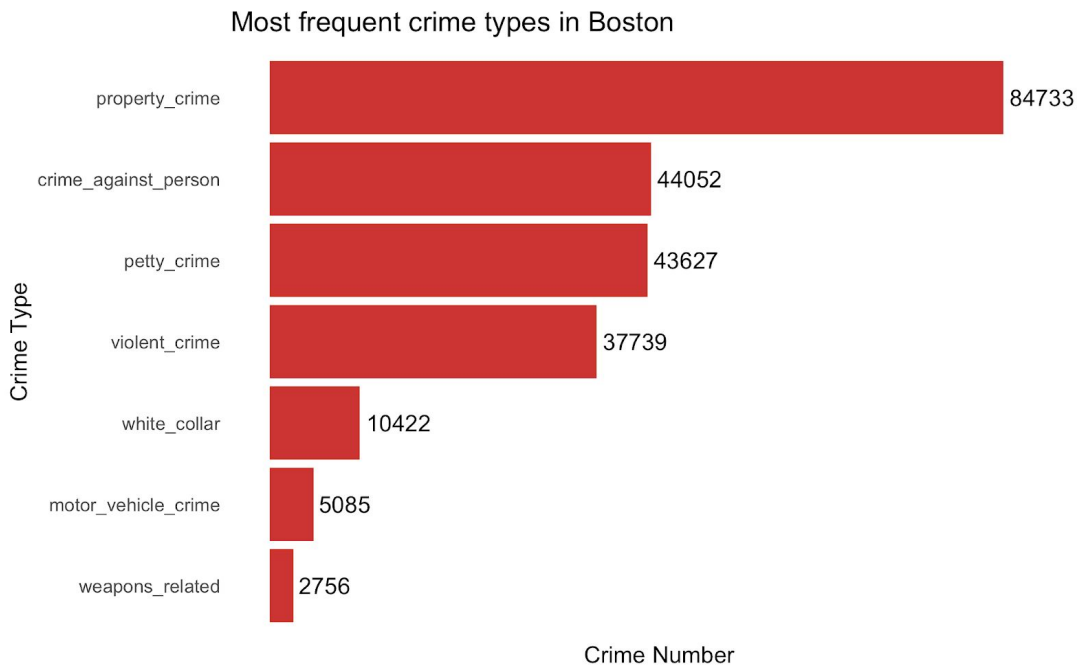
They were combined as follows:

Police District	Neighborhoods Covered
A-1, A-15	Downtown, Beacon Hill, Charlestown, Chinatown, North End, West End, Leather District, Bay Village, and the Downtown Waterfront.
A-7	East Boston
B-2	Roxbury/Mission Hill
B-3	Mattapan/North Dorchester
C-6	South Boston
C-11	Dorchester
D-4	Back Bay/South End/Fenway
D-14	Allston/Brighton
E-5	West Roxbury/Roslindale
E-13	Jamaica Plain
E-18	Hyde Park

Exploratory Data Analysis

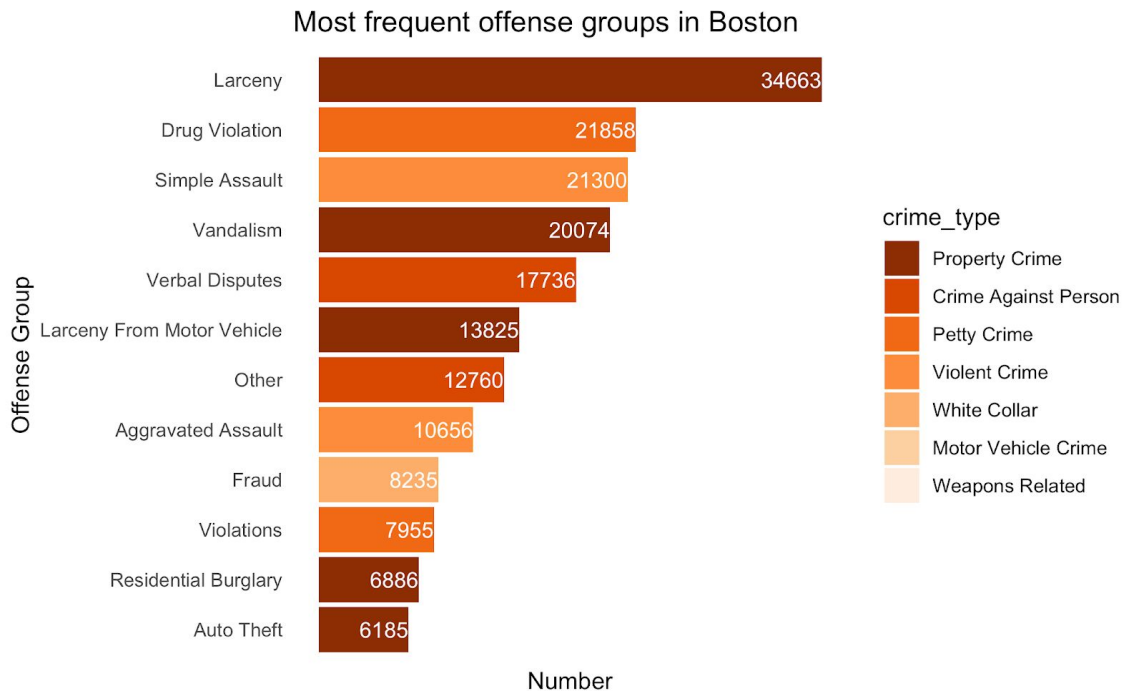
Part 1: General Situation

First, let's check the total numbers in each crime type over the entire time period in our dataset.



We see that property crime is the most common, and it is nearly twice as prevalent as the second most frequent crime type, crimes against a person. The second, third and fourth most common crime types, crimes against a person, petty crime and violent crime, are very similar in their total numbers. Note that there is a large difference in the crime numbers between the first four crime types and the last three: white collar crime, motor vehicle crime and weapons-related crimes. These crime types are far less common, but it is worth pointing out that these are subjective groupings of our own creation.

Accordingly, our exploration focuses on the four most common types of crime, which make up 92% of crime in Boston.

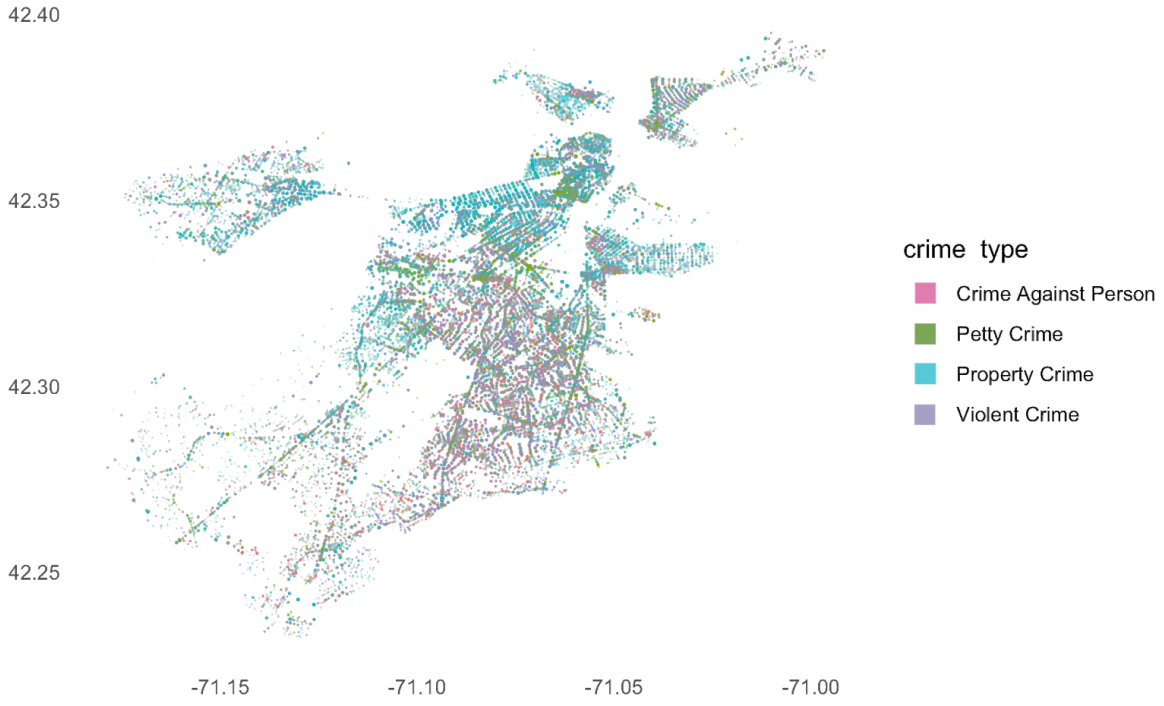


To better understand the groups, we'll examine the specific offenses within each crime type. Due to the high number of offense code groups, we'll focus on the top 12 most common.

We can see that Larceny is the most common offense group. Within the property crime group, larceny, vandalism, larceny from a motor vehicle, residential burglary and auto theft are most the most common. Within the crimes against a person group, verbal disputes and "other," which includes kidnapping, threats to do bodily harm, and intimidating a witness are the most common. Within the petty Crime group, Drug Violations are the most common, quite what we expected. Within the Violent Crime group, Simple Assault and Aggravated Assault are most common. Note that among the top 12 most common offense groups, Fraud is the only one that's not one of the 4 most common crime types.

In order to understand the general geographic distribution of different crime types, we plot the crime incidents using longitude as its x-axis and latitude as its y-axis.

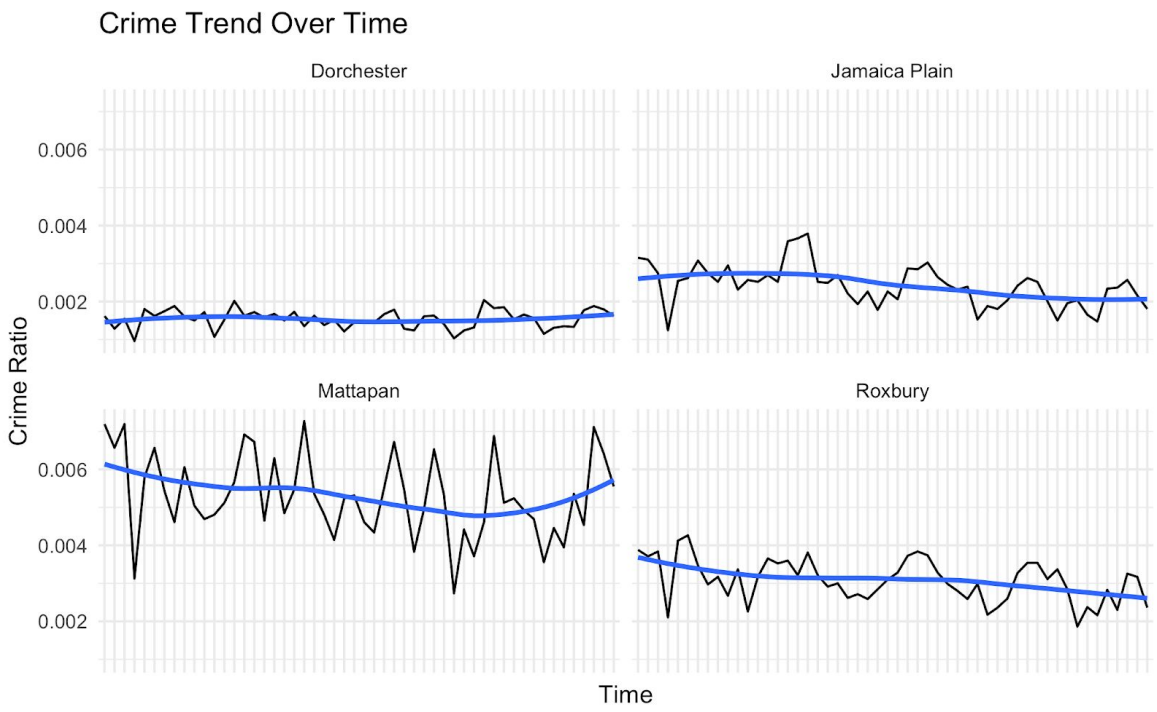
Observing from the graph, Most Property Crimes (blue dots) are at the Northeastern outskirts of Boston, while the rest three crime types are more mingled.



Also note that the crime density at the central part of Boston is much larger than that at the periphery. One possible reason is that the population density at central part of Boston is larger. As a result, in the exploration process followed, we divided the crime numbers by the population in each police district, so that we can compare the crime ratios across districts.

Part 2: Districts Exploration

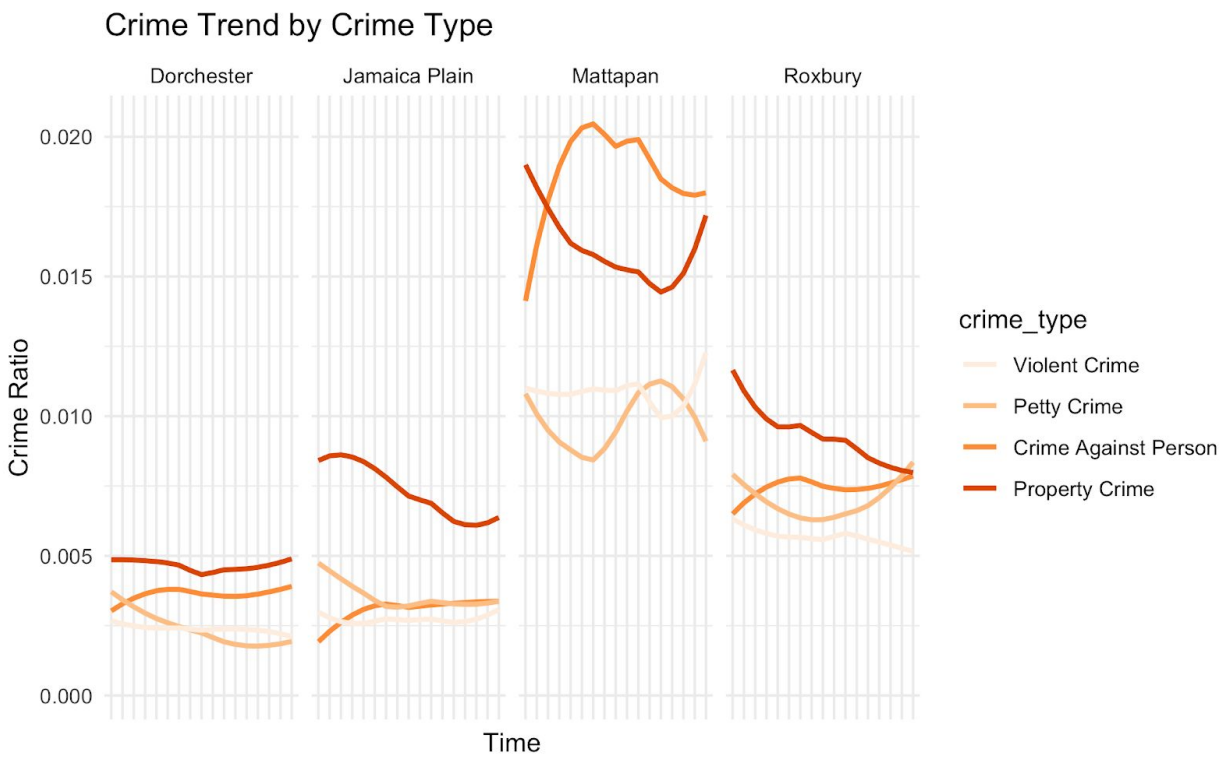
Now we shift our attention to the four districts selected, and try to explore how the crime ratios differ.



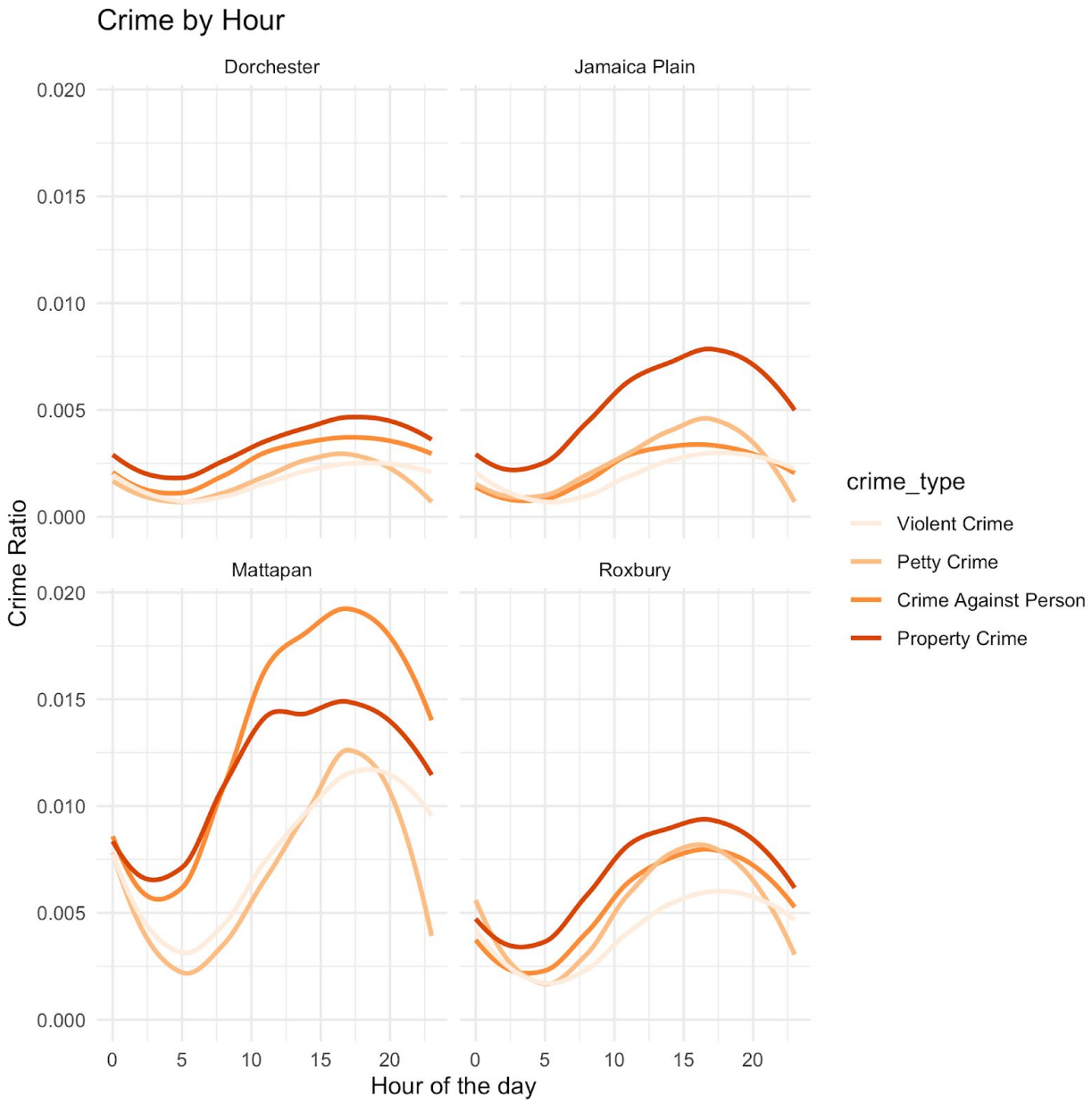
We first see how crime ratio changes over time in the four districts. Due to the fluctuations, we added the smooth lines to see the trends. From the graph, we have the following discoveries:

1. Mattapan always has the highest crime ratios and Dorchester always has the lowest crime ratios.
2. The crime ratio of Dorchester is flat over time, while there is a dip in Mattapan crime ratios, but it gradually rise back.
3. In contrast, the crime ratios in Jamaica Plain and Roxbury both have a decreasing trend over the time period.

The crime trends match our general expectations: Mattapan has more African and immigrant population and lower average income, so there is a higher crime ratio; Roxbury and Jamaica Plain both has a decreasing crime ratio trend as the two neighborhoods are undergoing gentrification processes.

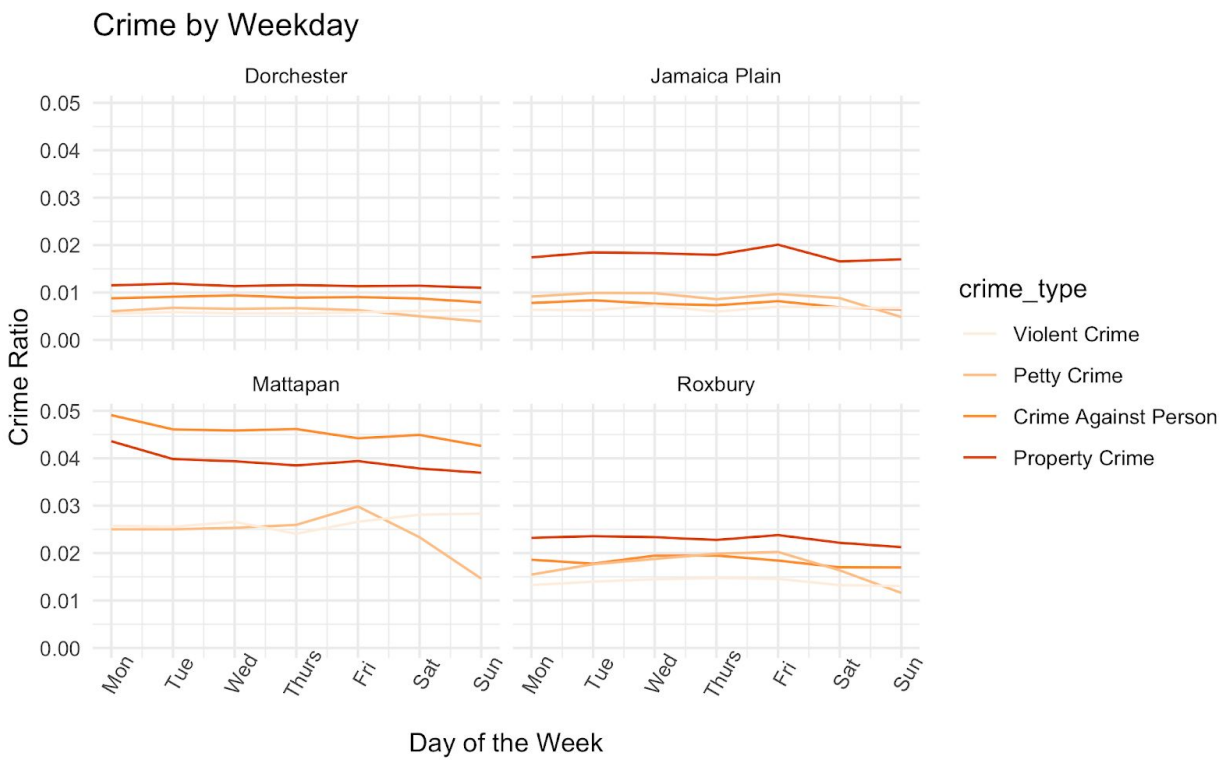


Then, we examine the crime trend among different crime types by plotting the smooth lines. We can see that the Property Crime, the red lines, have similar trends with the total crime ratio. Since the definition of gentrification is the influx of affluent residents and businesses in a neighborhood, the decreasing trend of property crime in Jamaica Plain and Roxbury can be explained by the process. On the other hand, we don't see any immediate relations in the change of other crime types and the gentrification process.



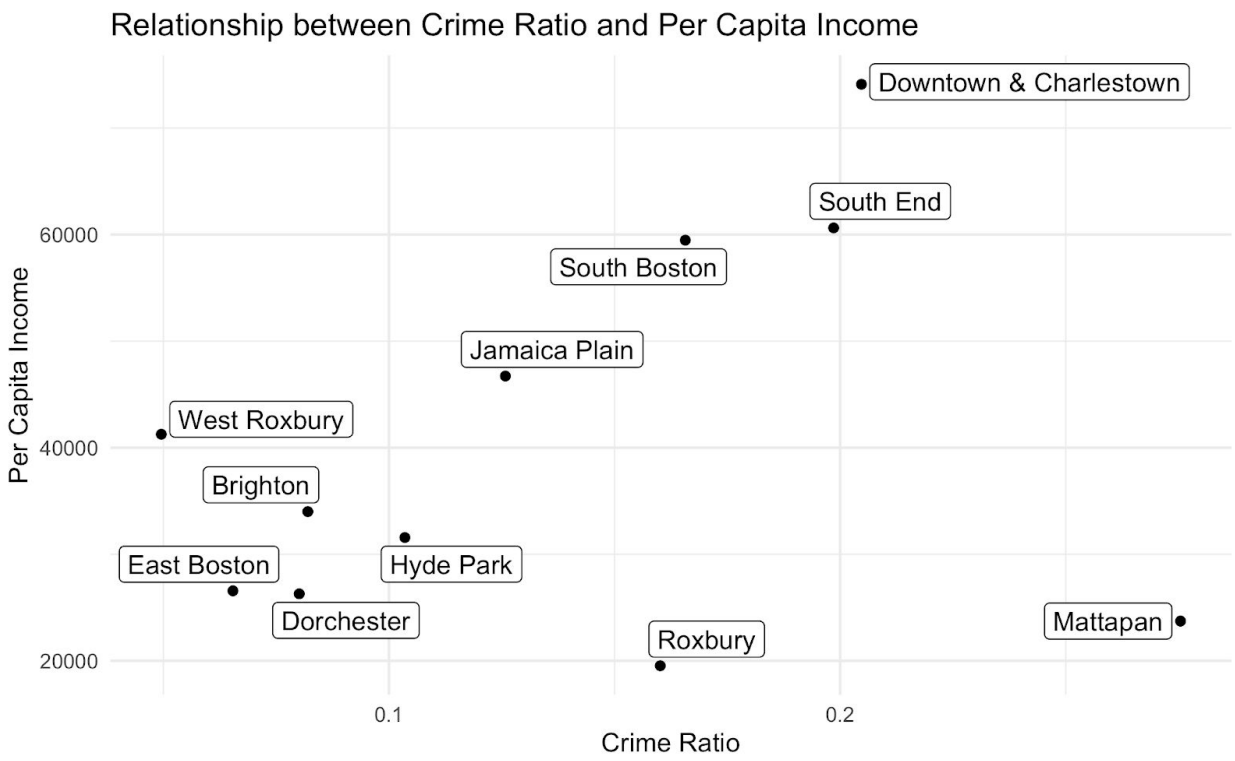
In this graph, we want to see how crime ratios change over the hour of the day from the smooth lines. In all districts, the minimum of crime ratios happen at around 5 a.m., when people are most asleep. Then, over the course of the day, the crime ratios gradually increase to the peak at around 17:30 p.m..

In addition, note that except in Mattapan, all other 3 districts have the highest crime ratios for Property Crime all the time. In Mattapan, Crime Against Person has higher crime ratio during most time of the day, where people are active and are more likely to get into conflicts.



Then, we want to examine the relationship between crime ratios and weekdays. The situations in the four districts is quite differently, so we will discuss them separately.

- For Dorchester, the variation of crime ratios between weekdays are very little.
- For Mattapan, Property Crime and Crime Against Person has similar trends, while Petty Crime peaks at Friday and decrease on weekends.
- For Jamaica Plain, Property Crime peaks at Friday.
- For Roxbury, there is an increasing trend in Petty Crime on weekdays, the crime ratio peaks at Friday then decrease over the weekend.



Next, we want to explore the relationship between Crime Ratio and Per Capita Income for each district. There seems to be a positive correlation, with one outlier Mattapan. However, we don't see the difference between gentrifying districts (Roxbury and Jamaica Plain) and other districts. Also, there are only 11 points (districts) on this graph, so they might not be sufficient to make any conclusions.

K-Means

We wanted to explore the possibility of a relationship between crimes and locations within districts or if they all are consistent within the district. To do this exploration we thought of using clustering algorithm as clustering number of crimes by latitude and longitude is easy to do and we can easily see the variability in crimes within districts. To do clustering we chose K-means clustering because K-means is a method of vector quantization and using this we can easily see the variability within districts if any.

What is K-Means?

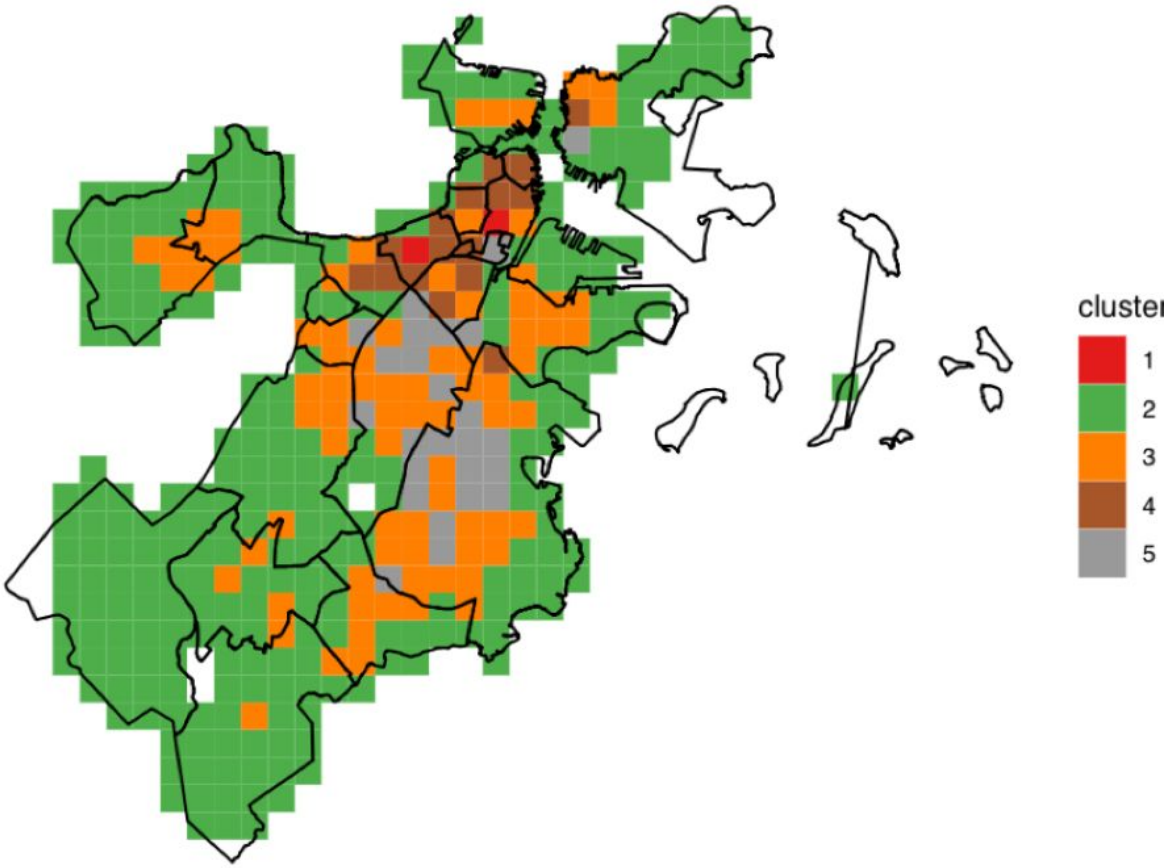
K-Means is one of the most popular "clustering" algorithms. K-means stores 'k' centroids that it uses to define clusters. A point is considered to be in a particular cluster if it is closer to that cluster's centroid than any other centroid.

K-Means finds the best centroids by alternating between

1. assigning data points to clusters based on the current centroids
2. Choosing centroids (points which are the center of a cluster) based on the current assignment of data points to clusters.[6]

Applying K-means is easy once we identify on what problem we want to cluster and what belief we are testing using this clustering technique. We applied K-means using latitude, longitude and total number of crimes. Using this approach, we were able to test our belief that Dorchester must be divided in clustering because it's a big district and it has a significantly larger population.

So, to use this technique we take latitude, longitude and crime reported in the Boston crime data and tried to cluster them. For this clustering we choose 5 clusters as in 5 cluster we were able to clearly see the difference between districts and within districts.



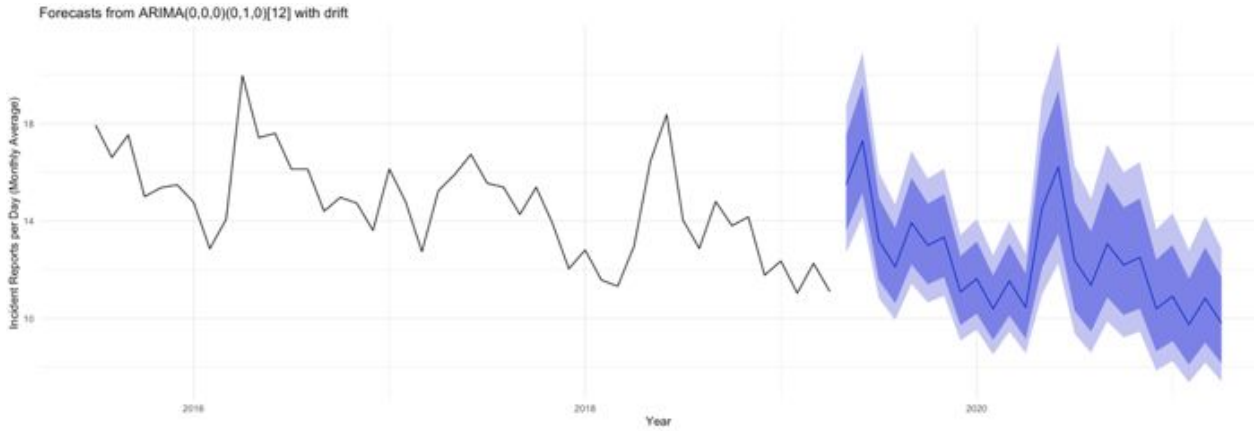
[6] https://stanford.edu/~cpiech/cs221/handouts/kmeans.html

To check for all crimes happened in past in Dorchester, we used time series plot to see the trend of crime happening in Dorchester and we used ARIMA series to predict the future crime trend in Dorchester.

Time-Series and ARIMA

A time series is a series of data points indexed (or listed or graphed) in time order. Most commonly, a time series is a sequence taken at successive equally spaced points in time. Thus, it is a sequence of discrete-time data.
We used time series to plot the crimes happening in Dorchester by month and used ARIMA to predict the future trend of crimes. **ARIMA** is an acronym that stands for Autoregressive Integrated Moving Average. It is a class of model that captures a suite of different standard temporal structures in time series data.

Dorchester Crime Prediction



We plotted this graph according to incident report per day and then taking its monthly average. We can see that claim by BPD is actually true that crime rate in Dorchester actually increased and we can see from the plot that in 2016 there were on average ~17 incidents per day and it decreased to ~11 incidents per day. There are seasonal trends too in which crimes increased but by looking at the graph we can say that overall it has decreased as compared to past years.

Roxbury

The Roxbury neighborhood is a hub of black culture in Boston. Located geographically in the very center of the city, its current population is over 50% black, and is home to several institutes dedicated to promoting the arts within the local community, including the Roxbury Center for Performing Arts, and the National Center of Afro-American Artists. At the core of the neighborhood's current identity is a rich history as a meeting and organizing place for blacks during and leading up to the American civil rights movement.

One of the notable demonstrations of the Civil Rights movements in the Northeastern United States was the March on Roxbury, Sept. 22, 1963. In response to the bombing of the 16th Street Baptist Church in Birmingham, AL, a week earlier, Boston's black leaders organized a march and presented the Boston School Committee with a proposal to end defacto segregation in Boston Public Schools.

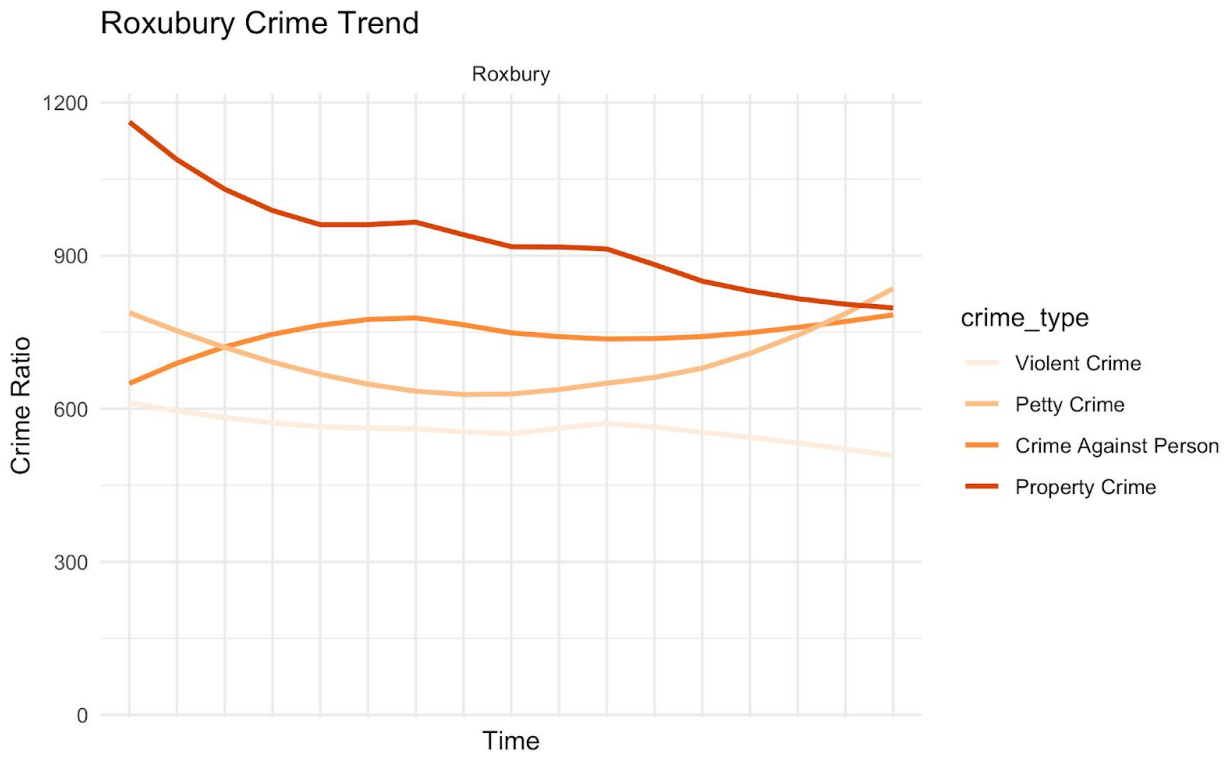
Roxbury certainly is responsible for shaping one of the seminal voices of the movement. At the age of 14, Malcolm X moved to Roxbury to live with his older half-sister, Ella Little-Collins, whom he credits for some of his early exposure to social activism. X lived in the neighborhood until he was 21.

Dr. Martin Luther King also spent time in the neighborhood while earning his PhD at Boston University. King would famously preach at the original location of the Twelfth Baptist Church on the corner of Shawmut Avenue and Melnea Cass Boulevard in Roxbury. The location of this church is notable, as it is the site of a cultural battle that started recently.

The Roxbury neighborhood is experiencing rapid gentrification. Many of the traditional residents are being pushed out, as developers and realtors scoop up land and property. One publication even went so far as to deem it "hyper gentrification." Further than the typical urban renewal associated with gentrification, there is an effort to essentially "rebrand" the neighborhood. The ground where Dr. King once addressed the congregation is now home to the Residence Inn by Marriott, Boston Downtown/South End. Another nearby hotel, The Alexandria, is undergoing a facelift renovation that has been officially classified as a development project in the South End.

Many Roxbury residents are upset by the changes and efforts to erase the culture from a place with such a proud heritage. The demographic changes appear to

happening too rapidly to be captured by publicly available data, however, we did notice a change in recent crime rates in the neighborhood.



Mattapan

"Mattapan" is claimed to be the original Native American name for the Dorchester area, as it started as a Native American community known as Mattahunt Tribe in the early 17th century. At the beginning of the 21st century, a significant number of Irish, Jewish, and Haitian immigrants settled down here and called it a "good place to sit". In 1870, Mattapan became part of the Boston together with neighboring Dorchester. In 2013 the population in Mattapan was 36,299. Of this total 11% were Caucasian, 82% were African American, 1% were Asian, 2% were a mixed race, and 6.5% were devoted to other races. Based on percentages in Mattapan, the cost of living is 8% lower than Boston, the total crime rate is 27% higher compared to Boston, the number of high school graduates are 11% lower than Boston, employment is 9% lower, and housing is 23% lower than Boston.

Future Work

Our exploratory work has led to just as many questions as answers. Luckily we have still been able to progress towards the truth we are seeking. Our work has allowed us to identify areas of interest for future research.

The dataset is still in its infancy with only approximately three years of data available. As more data is collected and released it will become far more valuable for research purposes. We have found that gentrification takes time, and only becomes apparent on closer to a five to ten year scale. While buildings go up quickly, people do not relocate immediately. It makes identifying areas of interest more difficult. Ideally we would be able to look at crime data and make decisions about the neighborhoods from there. At the moment it was much easier to look at neighborhoods of interest and explore how crime has been changing in those neighborhoods.

We would also like to be able to look at more data regarding the racial profile of the Boston police force. At the moment all that we could find was the racial makeup of the entire police force, not by department or neighborhood. As might be expected there is additionally no historical staffing data available. It made trying to identifying the significance of a representative police force difficult.

We have a hypothesis that making the police force representative of the underlying population would be beneficial in reducing race related arrests. Unfortunately at this point that has to be pure speculation. We believe that the data does exist, and if we could access it from the Boston police department it would prove invaluable to our exploration of how race affects crime in the city.

Specifically with the data we have, our research has pointed to further questions that can be explored without the need for more information. As shown when joining education, population and racial information to the crime data there is much more that can be seen regarding crimes in those neighborhoods. Unfortunately the education population and racial data is not as granular as the data available in the crime data set.

We were unable to control for underreported crimes. It was suggested that we use car theft as a reference point for crimes but at the time of completing this paper we were

unable to execute this exploration. There is ample research to suggest that all crimes are under-reported[7] and we hope that this would allow for us to more accurately track shifts in crime that are associated with gentrification.

Ethical Consideration

Ethical concerns include the potential misuse of the data for increased profiling of minorities. The intent is to better understand trends in crime in Boston and its potential relationship with gentrification but there is definitely room for inappropriate work.
It was important for us to focus on the socio-economic aspects associated with gentrification instead of race as this is more of a true connection. We do not believe anything covered in this report would constitute an ethical violation, nor likely anything that could be done with this dataset.
Additionally we saw issues with the lack of rape reporting. If the data exists there is potential that it could be controversial and potentially unethical to report it in more detail. But without access at this time, there is nothing unethical to report.

[7] **https://www.nap.edu/read/18605/chapter/5#38**

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